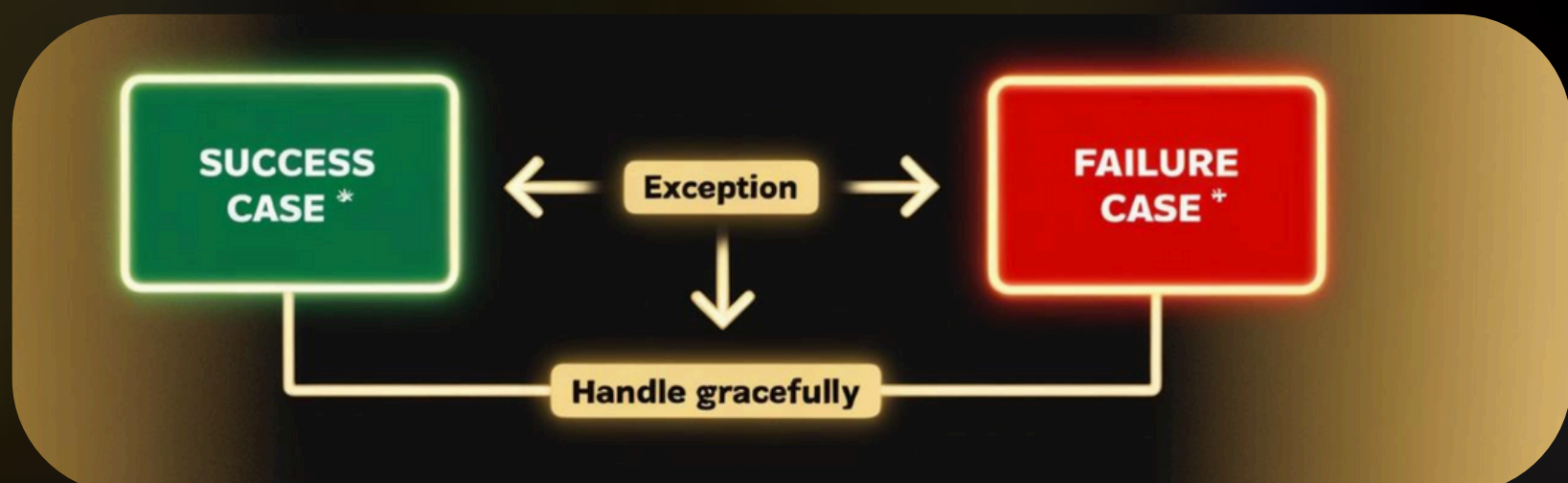




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WHAT ARE EXCEPTIONS IN JAVA



What are Exceptions in java

- Exception is an event that disturbs the normal flow of program execution.

What is Exception Handling in Java?

- Exception handling is a programming technique
- Used to handle the run time errors
- to maintain the normal flow of application.

Reason for Exception

Exception can be caused by many reasons

- When user enter invalid data.
- User try to open file, but file not found
- Server failure
- During communication network connection has been lost. And many reasons
- **However, well-written program should handle all possible exception.**

Why we need to handle Exception

- Once we deployed our Application into higher environment end user can access application
- And at that time user may face exception
- Exception can be caused by many reasons
- When user enter invalid data.
- User try to open file, but file not found
- Server failure
- During communication network connection has been lost. And many reasons
- However, well-written program should handle all possible exception
- And we should provide meaningful message instead to displaying stack trace to end user to end user to maintain normal flow of execution



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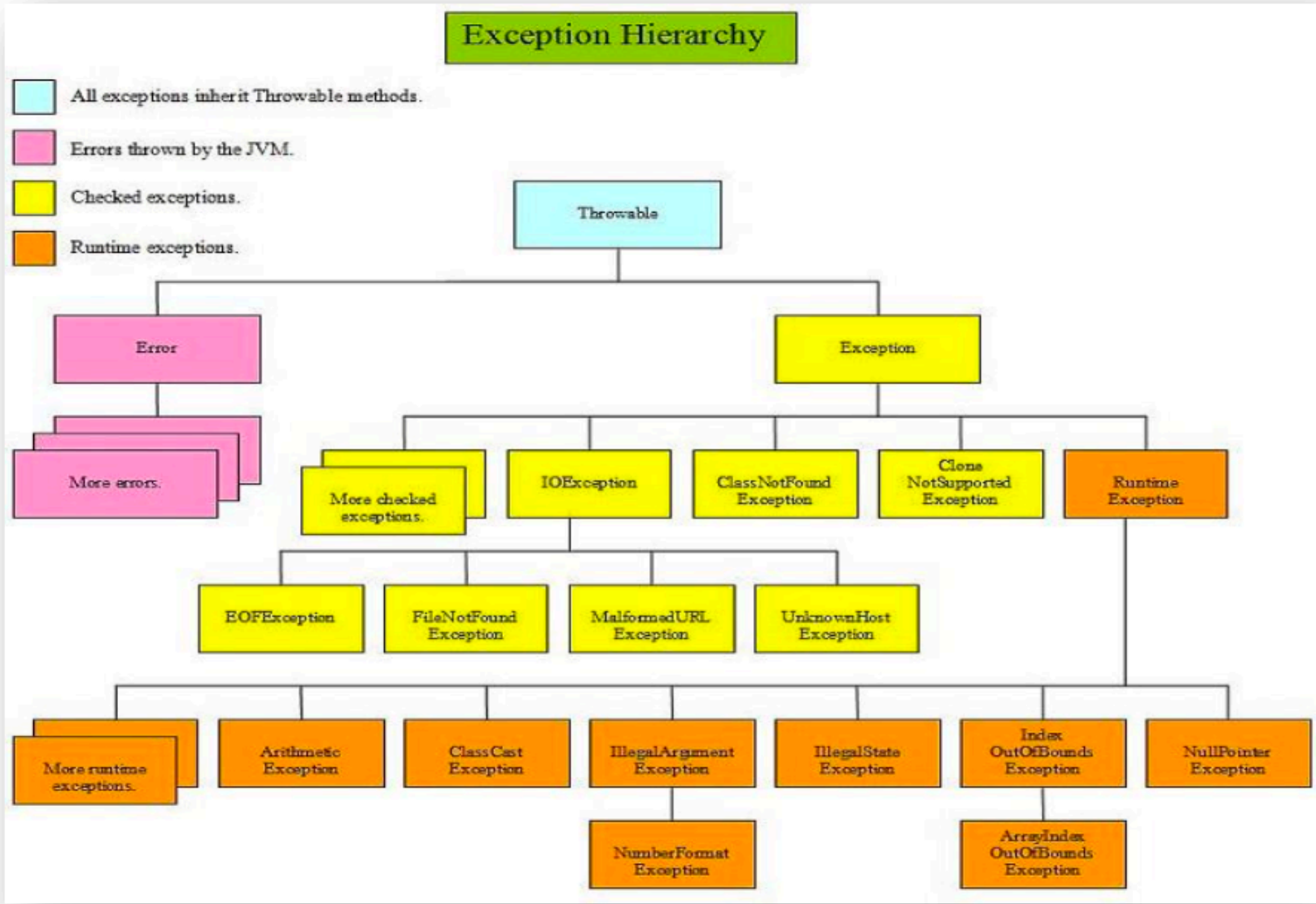


JAVA THROWABLE CLASS

Java Throwable class

- Super class of all types of error and exception in java
- All error types and exception types inherit the throwable class properties.
- JVM or throw statement Throws, only the instance of Throwable class.
- Throwable class defines the properties of objects that can be thrown by JVM or throw statement.
- Throwable class has some methods: used to get exception details

Method	Description
String getMessage()	Detail of exception description is returned.
void printStackTrace()	Detail of stack trace is returned.
String toString()	A short description is returned.





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ERROR CLASS

Error Class

- Error class represents the errors which are mainly caused by the environment in which application is running
- It is a part of java. lang package and Direct subclass of throwable class
- Recovering from error is not possible.
- Only solution to errors is to terminate the execution.
- All errors belong to only one category. i.e., Unchecked
- Compiler will not have any knowledge about unchecked exception it is not checked by compiler it occurs in run time

Example:

StackOverflowError:	Occurs when stack overflows
OutOfMemoryError:	occurs when JVM runs out of memory
VirtualMachimeError	

Exception Class

- Defined in java. lang package and Direct subclass of throwable class
- Exception class represents the exception which are mainly caused by application itself
- We can recover from exception using try-catch block
- By using try-catch block we can handle the Exception.
- We can maintain normal flow execution even if exception occur.

Exceptions in java divided into 2 categories: Checked and Unchecked

Checked Exception	Unchecked Exception
▪ checked Exception are checked at compile time by compiler ▪ It occurs at compile time	▪ Un-checked Exception are not checked at compile time ▪ Compiler do not have any knowledge about unchecked exception, it is not checked by compiler it occurs in run time
▪ The classes that are extends Throwable class, Except Runtime Exception and Errors are known as “Checked Exception” or caught Exception	▪ The classes that are extends RuntimeException class and Error class are known as “UnChecked Exception” or Uncaught Exception ▪ Errors are also called as UnChecked Exception” or Uncaught Exception. ▪ Unchecked exceptions inherit from the Error class or the RuntimeException class.
▪ It should be handled using try-or catch blocks or declared for propagation	▪ No need to handle or declared for propagation. ▪ Because it occurs in run time
Examples: ▪ IOException ▪ SQLException ▪ ClassNotFoundException ▪ FileNotFoundException ▪ InterruptedException ▪ CloneNotSuupotedException	RuntimeException class ▪ IllegalArgumentException. ▪ ArrayIndexOutOfBoundsException. ▪ ArithmeticException. ▪ NullPointerException. Error class ▪ StackOverflowError ▪ VirtualMachimeError ▪ OutOfMemoryError



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Error VS Exception

	Error	Exception
Related to	Error are mainly caused by environment of application	Exceptions mainly caused by Application itself
Recovery	Recovering from error is not possible	We can recover from exception.
Solution	The only solution to errors is to terminate the execution	By using try-catch block
try-catch block	- By using try-catch block we can't handle the errors. - Even if we handle errors using try-catch, our application will not recover from error.	- By using try-catch block we can able to handle the Exception. - And we can maintain normal flow of execution.
Categories	All errors belong to only one category. i.e., Unchecked.	- Exceptions in java divided into 2 categories. - Checked Exception. - Unchecked Exception.
Checked at:	Compiler will not have any knowledge about unchecked exception which include Errors and subclass of Runtime Exception because they occur at run time.	- Compiler will have knowledge about checked exception because they checked during compilation. - And compiler will force you to keep try-catch block when statements throws exception

throw v/s throws

throw	throws
Used to explicitly throw an exception.	Used to declare an exception.
- Throw keyword is mainly used to throw a Customized exception - By using throw keyword, we can explicitly throw an exception - We can throw either checked exception or unchecked exception	- throws keyword gives an information to the programmer about exception that might be occur during execution - It helps to programmer to provide an exception code to maintain normal flow of execution. - Concept: Predict exceptions that may throw during execution and provide code to handle exception
We can't throw more than one exception at a time.	We can declare more than one exception at a time using comma operator.
Throw keyword used with method	Throws keyword used with method declaration.
By using throw, we can't able to propagate the checked exception.	By using throw, we can propagate the checked exception

Example:
Void main()
{

throw new Eception1();

}
Exception: any Throwable type object.

Example:
Void main() throws Exception1,Exception2,,,,,
{

throw new Eception1();

}



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PURPOSE OF THIS DOCUMENT

- **Use this PDF as a quick last-minute revision guide only.**
- **For more details, please perform R&D and gain a deeper understanding of how memory architecture works.**
- **Once the interview series is complete, we will cover this topic in detail.**



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